PCBC H Parametrics Practice Test C Name _____ _____/12

Calculator | 30 minutes total for all four questions

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

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Signature _____

SHOW ALL WORK FOR FULL CREDIT!

Practice set: the calculator question comes first, as announced in class. Every question copies a homework problem with new numbers. 3 points each is a practice split of the posted 12.

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandth. *(practice instructions)*

1. The path of a projectile follows the parametric equations

$$x = v_0(\cos\theta)t \qquad \text{and} \qquad y = -16t^2 + v_0(\sin\theta)t + h_0$$

where θ is the angle of inclination of the projectile at the launch, v_0 is the initial velocity of the projectile in feet per second, and h_0 is the initial height of the projectile in feet.

A baseball player throws a baseball when it is 4 ft above the ground. The ball leaves his hand at an angle of 28° with respect to the ground at velocity 112 ft per second.

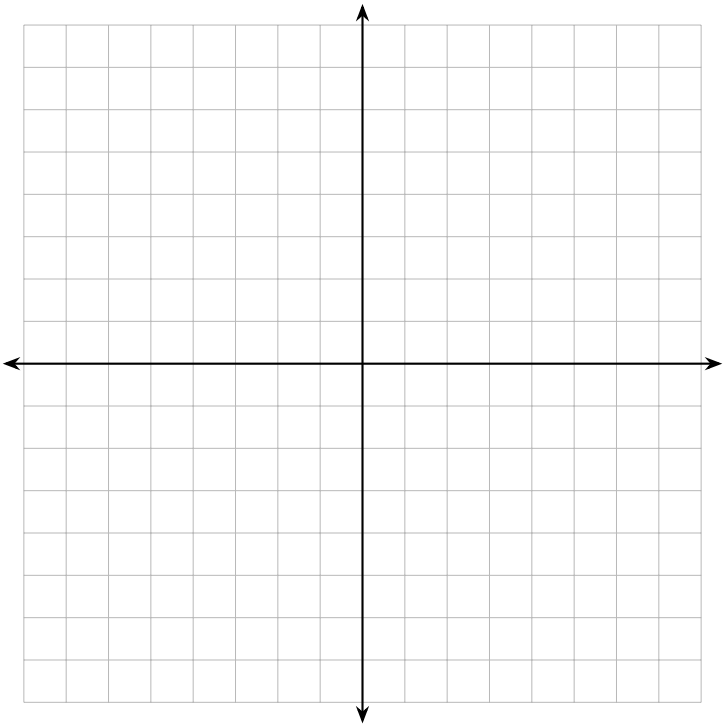
(a) Determine the parametric equations that model the path of the projectile. (b) Determine the rectangular equation that models the path of the projectile. (c) Determine approximately how long the projectile is in flight and the horizontal distance covered. **3 pts**

Non-Calculator 30 minutes total for all four questions

SHOW ALL WORK FOR FULL CREDIT!

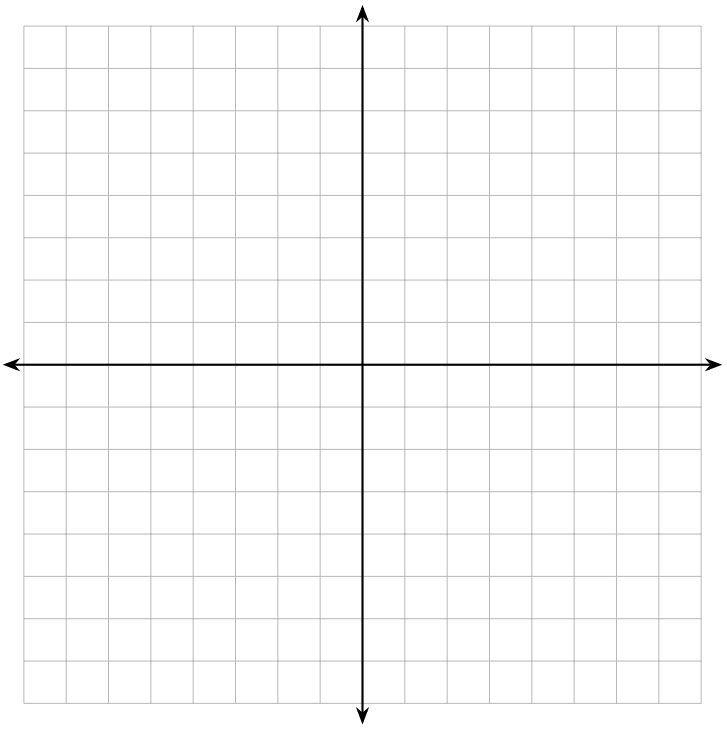
Graph the plane curve defined by the given parametric equations. Then find a rectangular equation for the curve. State any restrictions on the domain and show the direction of the curve. (3 points)

2. $x = 2t - 1, \quad y = t^2, \quad$ for t in $[-2, 2]$ **3 pts**



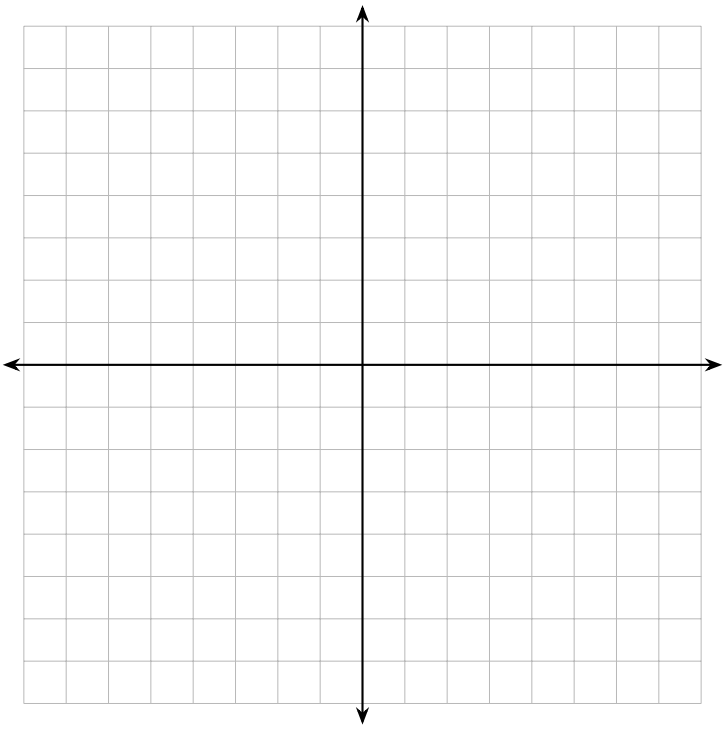
Find a rectangular equation for the given plane curve, and graph the curve. Show the direction of the curve. (3 points)

3. $x = -1 + 3\cos t, \quad y = 2 + 3\sin t, \quad 0 \le t \le 2\pi$ **3 pts**



Find a rectangular equation for the plane curve and graph the curve. State any restrictions on the domain. (3 points)

4. $x = t - 1, \quad y = \dfrac{2}{t-1}, \quad \text{for } t \neq 1$ **3 pts**



PCBC H Parametrics Practice Test C: Answer Key

12 points

Each solution follows the method used in class. The gray box shows another way that also works. Once t is gone, any equivalent form of the answer is fine. Every value was checked in Python.

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1. *Calculator* (like homework day 2 #19)

(a) $\boxed{x = (112\cos 28^\circ)t, \quad y = -16t^2 + (112\sin 28^\circ)t + 4}$

(b) $\boxed{y = 4 + (\tan 28^\circ)x - \dfrac{x^2}{784\cos^2 28^\circ}}$ (c) $\boxed{\approx 3.361 \text{ sec}}$, $\boxed{\approx 332.339 \text{ ft}}$

In class	Why
(a) $x = (112\cos 28^\circ)t,\ y = -16t^2 + (112\sin 28^\circ)t + 4$	Fill in the formula.
(b) $t = \dfrac{x}{112\cos 28^\circ}$	Solve the x equation for t.
$y = -16\left(\dfrac{x}{112\cos 28^\circ}\right)^2 + (112\sin 28^\circ)\dfrac{x}{112\cos 28^\circ} + 4$	Plug into y.
$y = 4 + (\tan 28^\circ)x - \dfrac{x^2}{784\cos^2 28^\circ}$	$\frac{16}{112^2} = \frac{1}{784}$, $\frac{\sin}{\cos} = \tan$.
(c) Graph $x1(t)$, $y1(t)$; trace to $y = 0$; set the trace step to 0.01, then 0.001, until y is as close to 0 as you can get	Degree mode.
Read $t \approx 3.361$ and $x \approx 332.339$	Time, then distance.

Other way. (c) Quadratic formula on $0 = -16t^2 + (112\sin 28^\circ)t + 4$ gives $t \approx 3.361$; then $x = (112\cos 28^\circ)(3.36069\ldots) \approx 332.339$. Or graph the rectangular equation and find its positive x-intercept.

Practice scoring (3 pts): 1 for the parametric equations, 1 for the rectangular equation, 1 for the time and distance.

Trap: rounding t to 3.36 first (≈ 332.271), or radian mode.

2. $x = 2t - 1, \quad y = t^2, \quad$ for t in $[-2, 2]$ *Non-Calculator* (like homework day 1 #3 and #4)

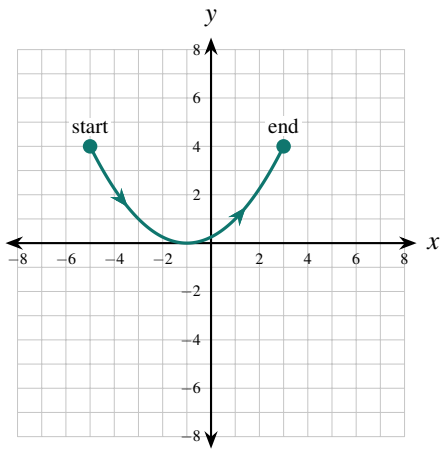
$\boxed{y = \left(\frac{x+1}{2}\right)^2, \text{ for } x \text{ in } [-5, 3]}$

From $(-5, 4)$ to $(3, 4)$, solid dots, no arrows on the ends.

In class	Why
t-chart: $t = -2, -1, 0, 1, 2$	Brackets: both ends in.
Plot, connect, arrows left to right	Lowest t to highest t.
$t = \frac{x+1}{2}$	Pick x: one answer. ($y = t^2$ would give $\pm$.)
$y = \left(\frac{x+1}{2}\right)^2$	Substitute. No need to multiply out.
for x in $[-5, 3]$	Min and max of the x column.

t	(x, y)
-2	$(-5, 4)$
-1	$(-3, 1)$
0	$(-1, 0)$
1	$(1, 1)$
2	$(3, 4)$

Closed interval: it stops at both ends.



Other way. Multiplied out it's $y = \frac{x^2+2x+1}{4}$. Check the graph on the calculator: Parametric, $-2 \le t \le 2$.

Practice scoring (3 pts): 1 for the equation, 1 for x in $[-5, 3]$, 1 for the graph with end dots and direction.

Trap: the whole parabola, or arrows on the ends.

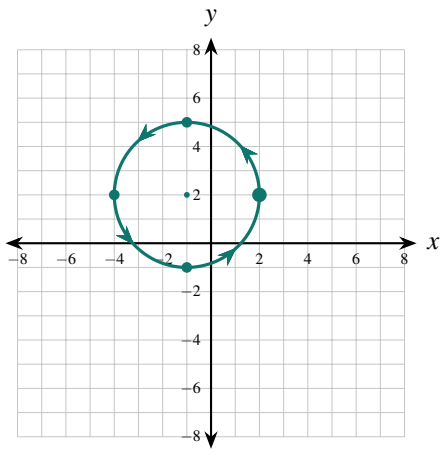
3. $x = -1 + 3\cos t, \quad y = 2 + 3\sin t, \quad 0 \le t \le 2\pi$ *Non-Calculator* (like homework day 2 #10)

$\boxed{(x+1)^2 + (y-2)^2 = 9}$
Whole circle, center $(-1, 2)$, radius 3, counterclockwise from $(2, 2)$. No domain needed.

In class	Why
t-chart: $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$	Quadrant values.
Plot, connect, arrows: $(2,2)$ to $(-1,5)$	Right, then top: counterclockwise.
$\cos t = \dfrac{x+1}{3}, \quad \sin t = \dfrac{y-2}{3}$	Isolate the trig.
$\dfrac{(x+1)^2}{9} + \dfrac{(y-2)^2}{9} = 1$	Square, $\sin^2 t + \cos^2 t = 1$.
$(x+1)^2 + (y-2)^2 = 9$	Times 9.

t	(x, y)
0	$(2, 2)$
$\frac{\pi}{2}$	$(-1, 5)$
π	$(-4, 2)$
$\frac{3\pi}{2}$	$(-1, -1)$
2π	$(2, 2)$

$t = 0$ and 2π land on the same point: the circle closes.



Other way. Read it off: the added numbers are the center $(-1, 2)$, the number in front of cos and sin is the radius 3.

Practice scoring (3 pts): 1 for the equation, 1 for center $(-1, 2)$ and radius 3, 1 for counterclockwise arrows.
Trap: center $(1, -2)$. $x = -1 + 3\cos t$ means $x + 1$.

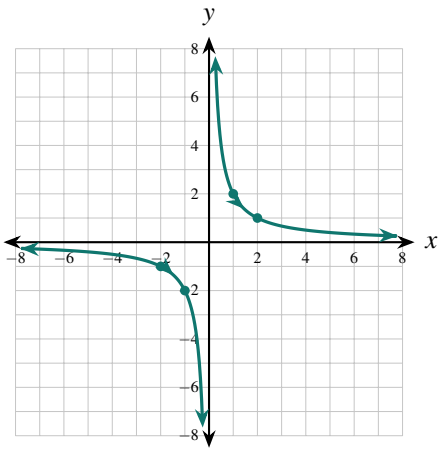
4. $x = t - 1, \quad y = \dfrac{2}{t-1}, \quad \text{for } t \neq 1$ *Non-Calculator* (like homework day 2 #11)

$\boxed{y = \dfrac{2}{x}, \text{ for } x \neq 0}$
Both branches, each traced left to right, arrowheads on every end.

In class	Why
$x = t - 1$ and $y = \dfrac{2}{t-1}$	Spot that $t - 1$ is in both.
$y = \dfrac{2}{x}$	Replace $t - 1$ with x.
t-chart: $t = -1, 0, 2, 3$	Skip $t = 1$; pick both sides.
Plot both branches, arrows to the right	x grows as t grows.
for $x \neq 0$	$t \neq 1$ carries over (good practice).

t	(x, y)
-1	$(-2, -1)$
0	$(-1, -2)$
2	$(1, 2)$
3	$(2, 1)$

Asymptotes $x = 0$ and $y = 0$. Both branches keep going.



Other way. Isolate t the long way: $t = x + 1$, so $y = \frac{2}{(x+1)-1} = \frac{2}{x}$.

Practice scoring (3 pts): 1 for $y = \frac{2}{x}$, 1 for $x \neq 0$, 1 for both branches with arrows.
Trap: only one branch. t can be less than 1 or more than 1.